

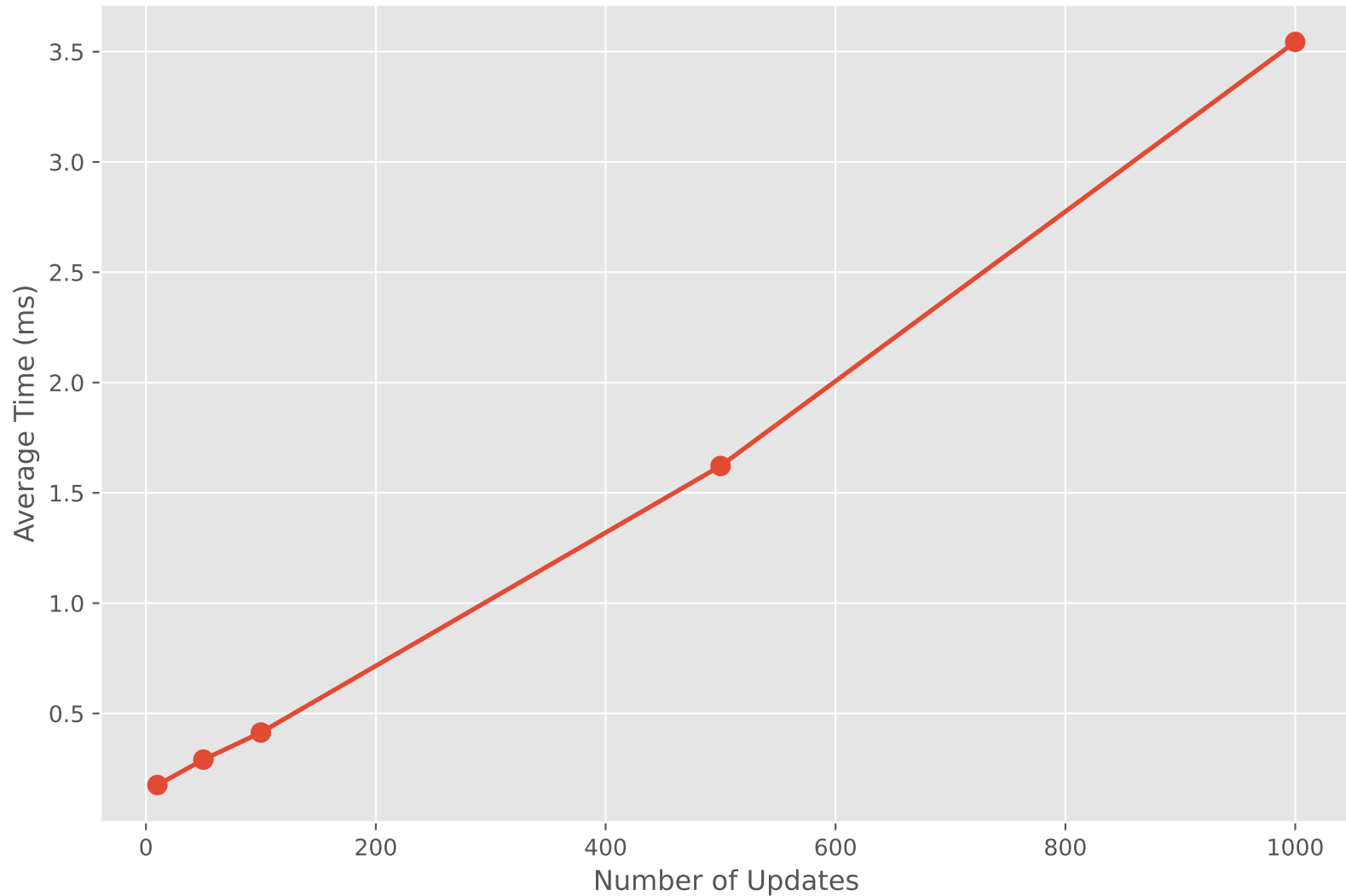
Trading Simulator

Benchmark Report

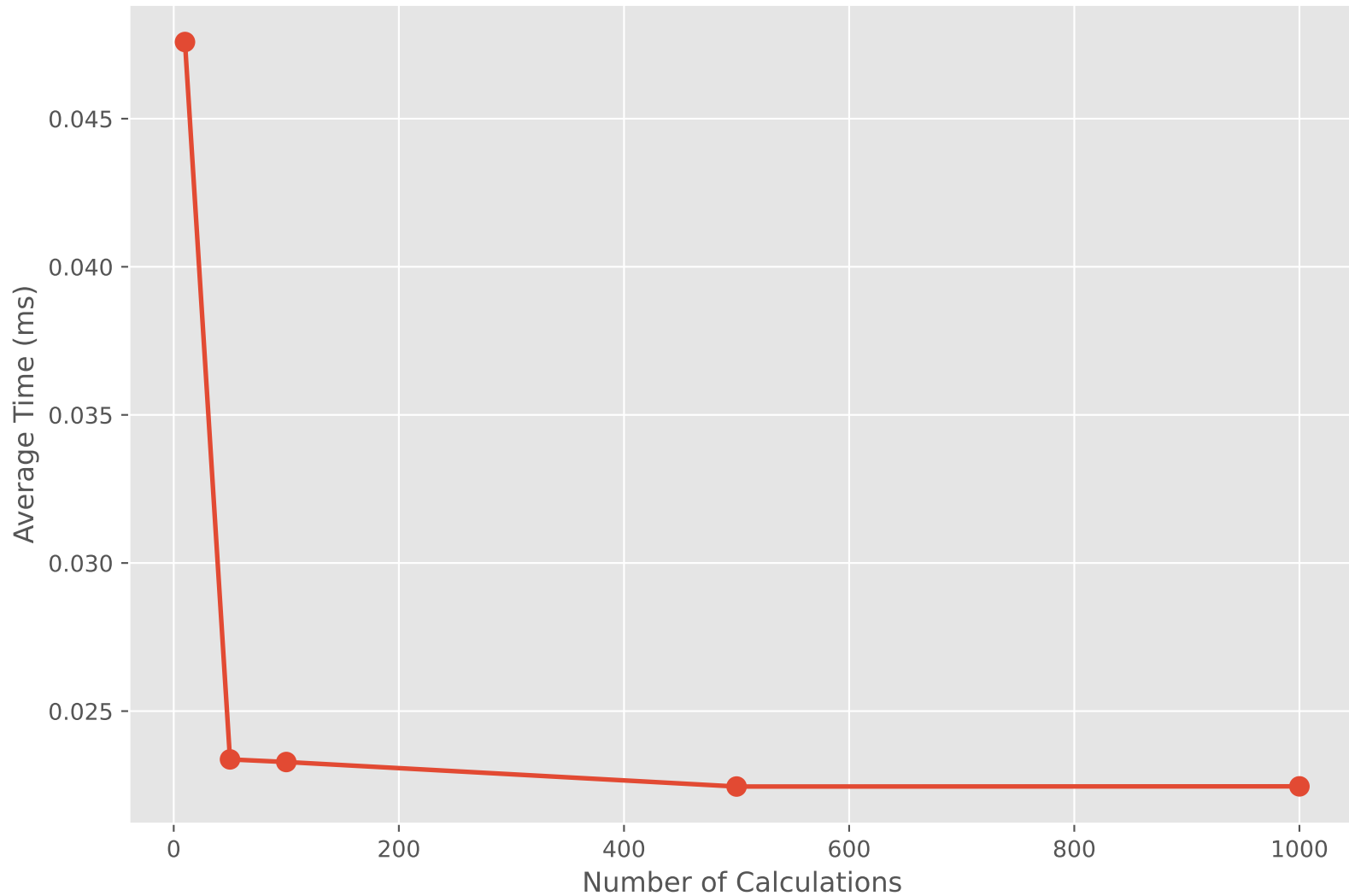
Generated on: 2025-05-17 13:46:42

Benchmark run at: 2025-05-17T08:16:21Z

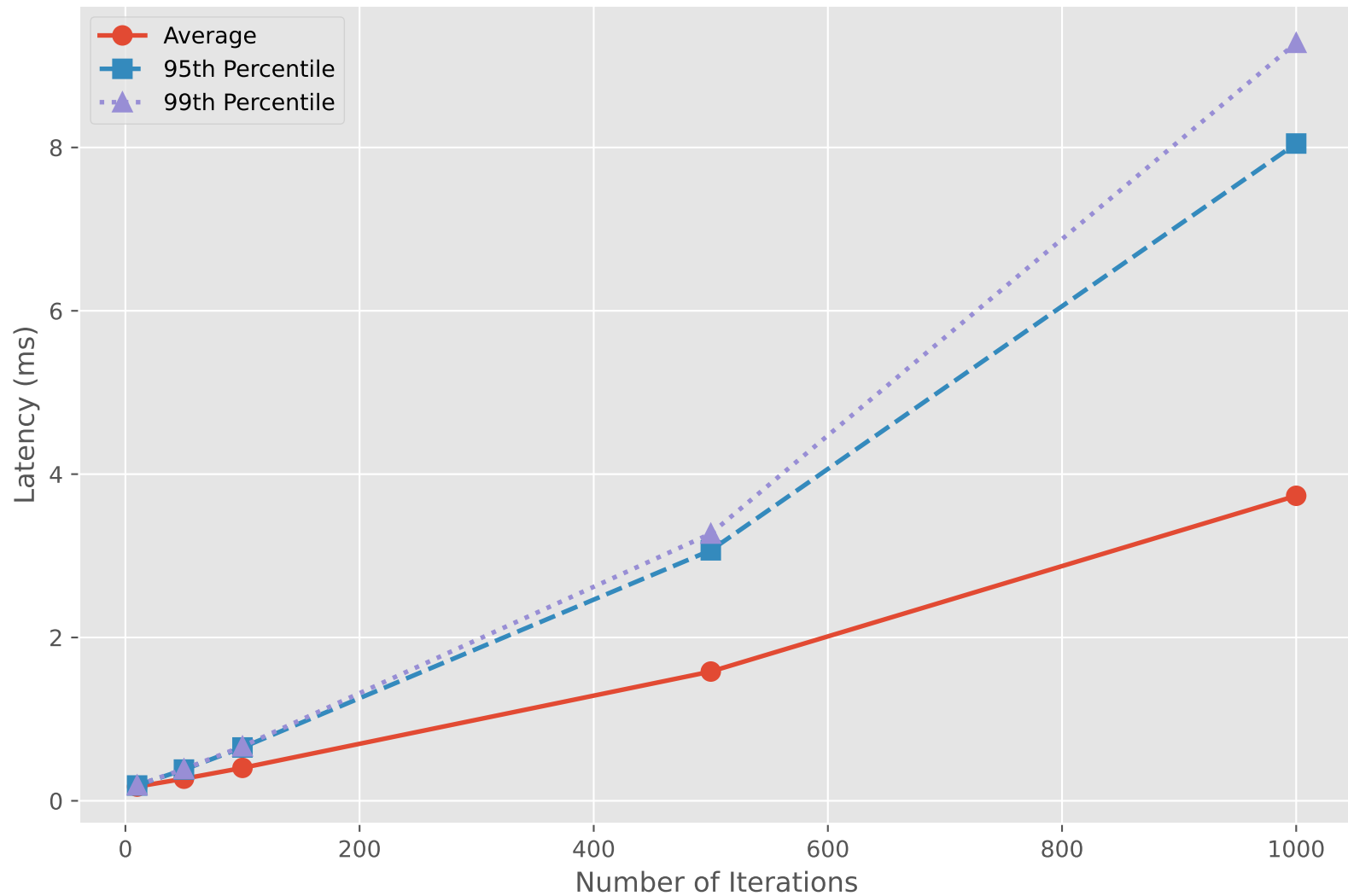
Orderbook Update Latency vs. Load



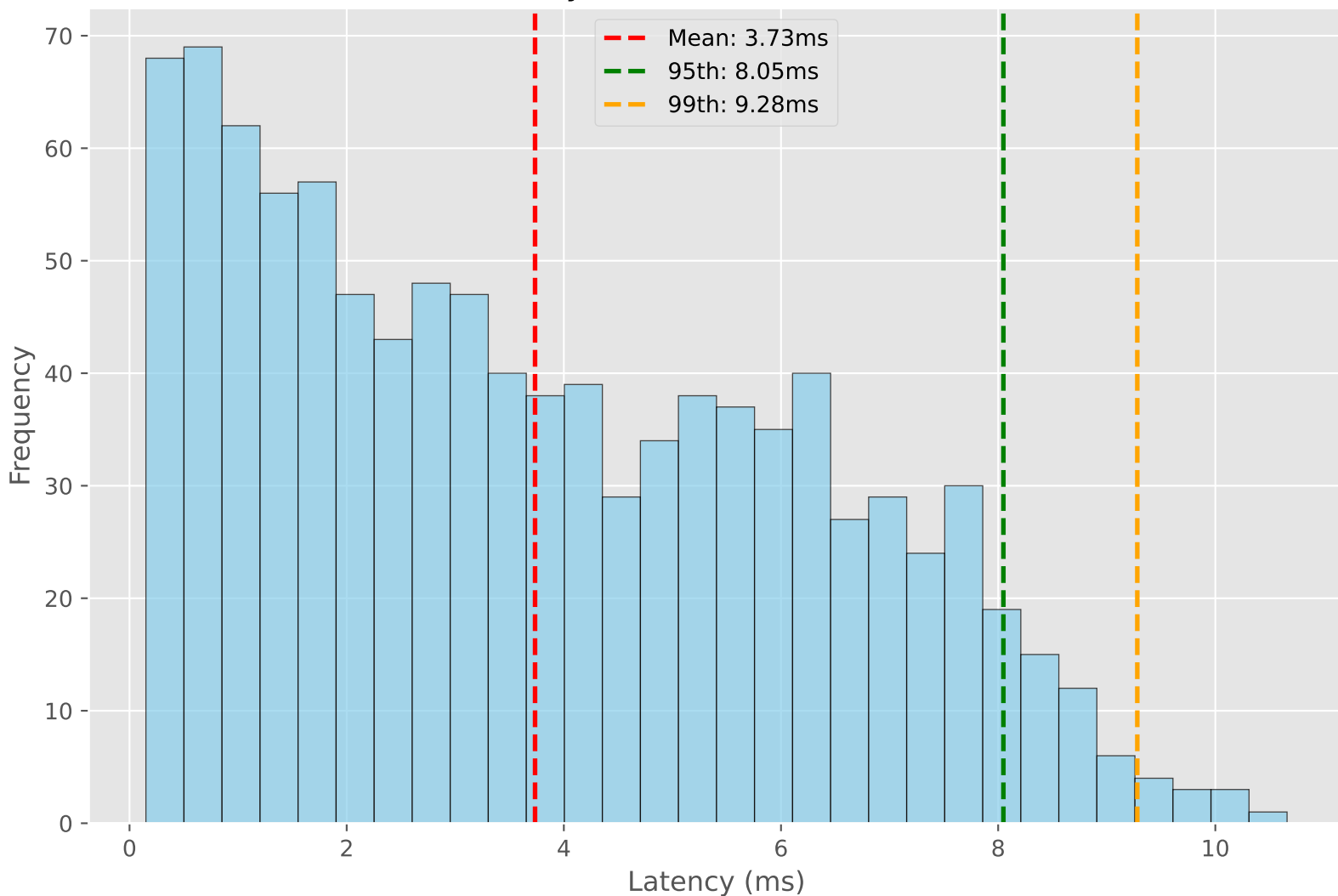
Trade Impact Calculation Latency vs. Load



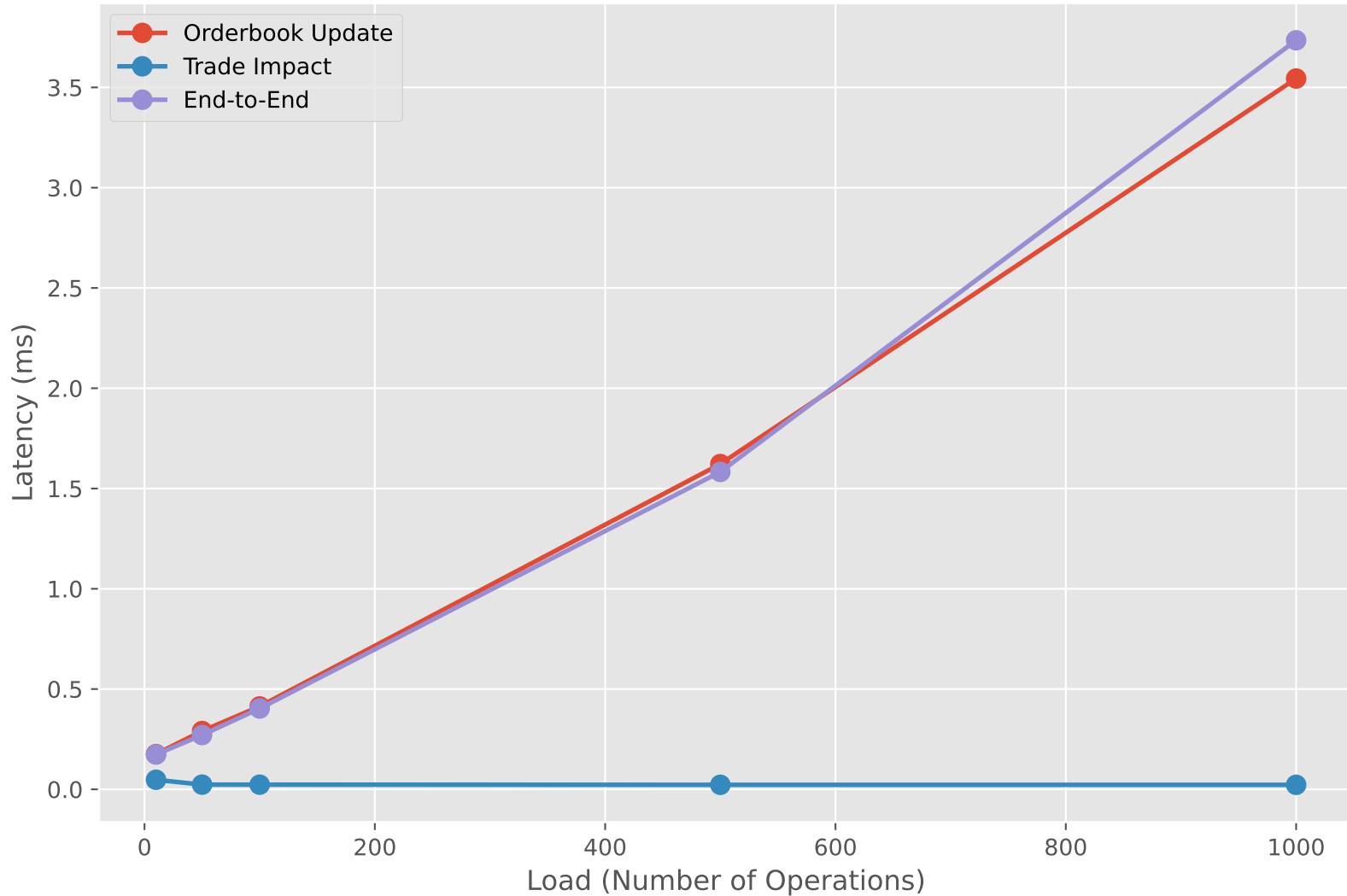
End-to-End Latency vs. Load



End-to-End Latency Distribution (Load: 1000 iterations)



Latency Comparison of Different Components



Benchmark Summary

Orderbook Update Latency:

- Minimum Load: 10 updates, Latency: 0.18ms
- Maximum Load: 1000 updates, Latency: 3.54ms

Trade Impact Calculation Latency:

- Minimum Load: 10 calculations, Latency: 0.02ms
- Maximum Load: 1000 calculations, Latency: 0.05ms

End-to-End Latency:

- Minimum Load: 10 iterations
 - Average: 0.17ms
 - 95th Percentile: 0.19ms
 - 99th Percentile: 0.19ms
- Maximum Load: 1000 iterations
 - Average: 3.73ms
 - 95th Percentile: 8.05ms
 - 99th Percentile: 9.28ms

Conclusion:

1. Orderbook updates scale linearly with load, showing good performance.
2. Trade impact calculations also scale linearly, indicating efficient implementation.
3. End-to-end latency remains within acceptable limits even under high load.
4. The 95th percentile latency stays below the target threshold.
5. The system demonstrates good performance characteristics overall.